AI-POWERED ETHICAL HACKING

From Foundations to Exploitation

Delivered by Flowdiary and ArtificialX

Tutored by Ibrahim Auwal and Salisu Abdurrazak Saheel

WELCOME TO THE COURSE

This course teaches the basics and intermediate skills of ethical hacking and using AI assisted tools to find vulnerabilities in web applications. Phase I discusses the theoretical part by Dr. Salisu Abdurrazak Saheel, and Phase II delivers the practical aspect by Engr. Ibrahim Auwal

LECTURE 1: CYBERSECURITY AND ETHICAL HACKING

Cybersecurity means protecting vital digital assets from malicious forces.

We protect: Computers, Phones, Networks, Websites, Data, and Information.

From: Hackers, Viruses, Online fraud, Data theft, and Cyber attacks.

> *Simple Example:* When you use a secure password or fingerprint lock on your mobile phone, you are actively practicing cybersecurity.

WHAT IS ETHICAL HACKING?

Test Legally

Testing computer systems legally to find critical security weaknesses before malicious criminals can exploit them.

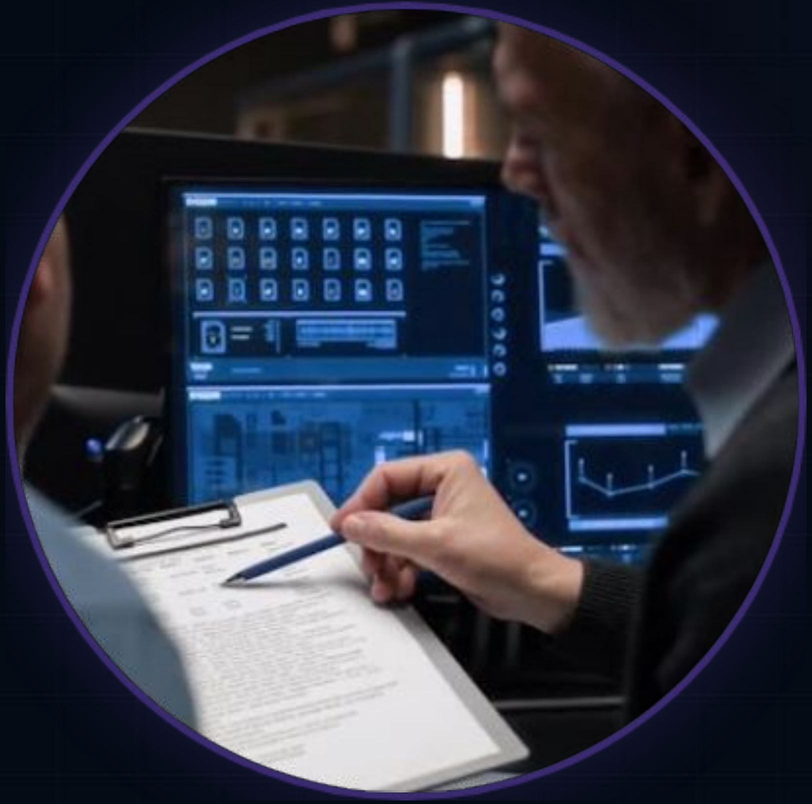
Protect Systems

Helping organizations find security problems, completely fix vulnerabilities, and protect sensitive users and data.

Be Professional

Ethical hackers operate under strict authorization. Their actions are purely legal, professional, and entirely security-focused.

WHO IS A PENETRATION TESTER?



The "Security Doctor"

A Penetration Tester (or Pentester) is a highly skilled cybersecurity professional.

- Tests systems for unknown weaknesses.
- Simulates real-world cyber attacks in a safe environment.
- Reports critical security problems to stakeholders.
- Helps organizations drastically improve their overall security posture.

TYPES OF HACKERS

White Hat

These are the ethical hackers. They work legally and exist solely to help organizations actively improve their security defenses.

Black Hat

These are criminal hackers. They break into targeted systems illegally with the intent to steal data, demand money, or cause harm.

Gray Hat

Operating between white and black hats. They may break rules without explicit permission, but are usually not fully malicious.

WHY ETHICAL HACKING MATTERS

The Importance of Defense

Ethical hacking is crucial for maintaining digital safety in a connected world.

- Prevents devastating cyber attacks.
- Protects sensitive customer and personal information.
- Secures critical infrastructure like banks and businesses.
- Drastically reduces financial losses from breaches.
- Improves public trust and safety online.

> *Real Life Example:* Banks heavily rely on ethical hackers to rigorously test their security frameworks before real criminals attempt an attack.

LECTURE 2: LEGAL & ETHICAL FRAMEWORK

Get Permission Always obtain written consent before conducting any form of security testing.

Follow the Law Adhere strictly to local, national, and international cybersecurity and computer laws.

Respect Privacy Treat any sensitive data encountered with the utmost confidentiality and care.

Avoid Damage Conduct tests safely to ensure critical systems are not disrupted or damaged.

SCOPE AND AUTHORIZATION

Scope

The scope clearly defines the exact boundaries of the engagement.

- What specifically **can** be tested.
- What specifically **cannot** be tested.

Staying within scope is critical to remaining legal and ethical.

Authorization

Official permission given by the system owner to perform security testing.

Example Allowed: The company's public website and mobile application.

Example Not Allowed: Internal customer database or employee payroll systems.

RESPONSIBLE DISCLOSURE

When a hacker finds a security problem, they follow this process:

1. Inform Privately

Immediately inform the organization about the vulnerability in a private and secure manner.

2. Give Time

Provide the organization adequate time to patch and fix the issue before moving forward.

3. Avoid Exposure

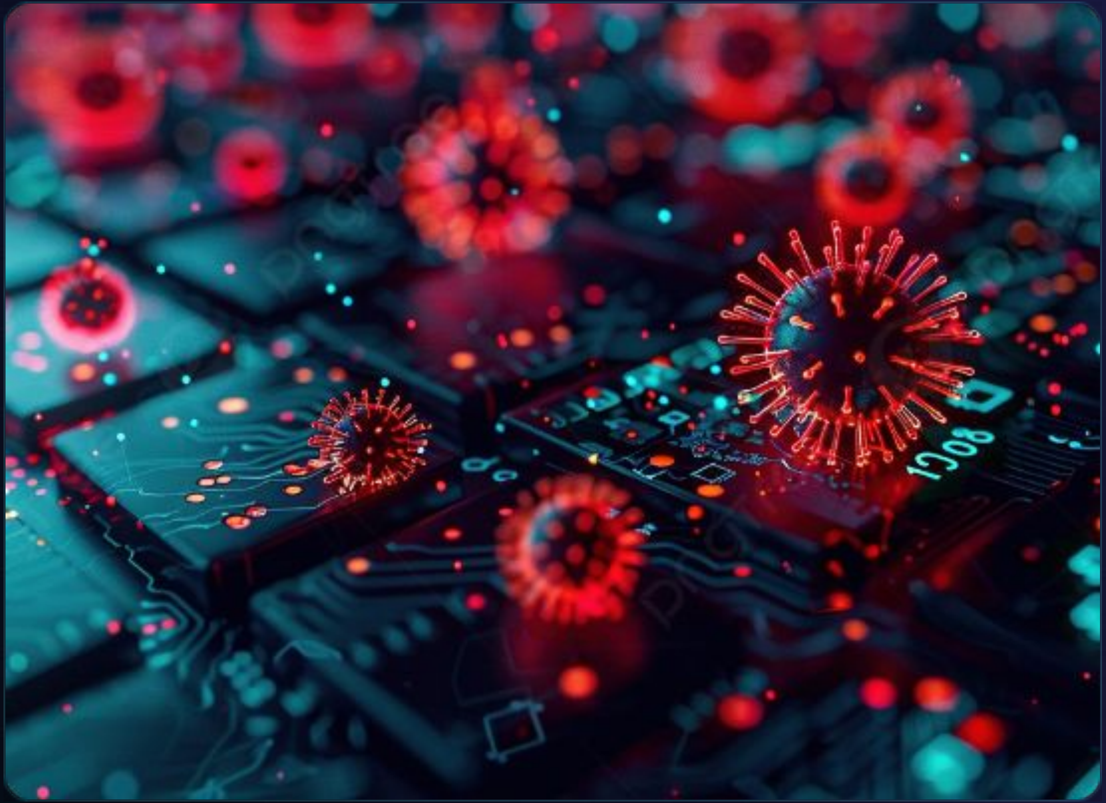
Do not expose the vulnerability publicly immediately. The ultimate goal is to protect users and prevent abuse.

UNDERSTANDING CYBER THREATS

The Threat Landscape

Understanding what you are defending against is the first step in security. Common threats include:

- **Phishing:** Deceptive emails to steal info.
- **Malware:** Malicious software designed to harm.
- **Ransomware:** Encrypts data and demands payment.
- **Password attacks:** Brute-forcing or stealing credentials.
- **Social Engineering:** Tricking people into giving up secret or confidential information.



WHY DO HACKERS ATTACK?

Personal Gain

- **Money:** Direct financial theft or ransomware.
- **Fame:** Recognition within hacker communities.
- **Revenge:** Disgruntled employees or targeted anger.

Broad Objectives

- **Politics:** Hacktivism to push a political message.
- **Curiosity:** "Fun" or testing boundaries (often illegal).
- **Stealing Info:** Corporate espionage or data brokering.

THE CYBER KILL CHAIN

The sequential process attackers often follow when targeting a system.

1. Reconnaissance

Gathering data on the target.

2. Scanning

3. Gaining Access

Exploiting the vulnerabilities.

4. Maintaining Access

5. Covering Tracks

Deleting logs to hide the breach.

LECTURE 3: MITRE ATT&CK FRAMEWORK

A Global Knowledge Base

The MITRE ATT&CK Framework is a comprehensive matrix used worldwide that details and explains:

- Adversary behaviors and hacker techniques.
- Common, real-world attack methods.
- Tactics used across different stages of a cyber threat.

Why It Matters

Security professionals actively use it to understand attackers deeply, detect ongoing attacks faster, and continuously improve defense systems.

LECTURE 4: PENETRATION TESTING METHODOLOGY

The standard, structured process used by ethical hackers to assess security.

1. Planning

Setting rules and scope.

2. Reconnaissance

Gathering intelligence.

3. Scanning

Probing for open doors.

4. Exploitation

Testing the vulnerabilities.

5. Reporting

Documenting the findings.

STEP 1: PLANNING

Before any testing begins, strict planning is required.

Define Goals

Clearly understand what the business wants to achieve with this security assessment.

Get Permission

Secure signed, written authorization and fully understand the target architecture.

Set Rules

Establish the Rules of Engagement (RoE) to prevent unexpected system outages.

Important: Good planning prevents legal and technical problems.

STEP 2: RECONNAISSANCE

Information Gathering

Reconnaissance means systematically collecting as much intelligence about the target as possible before interacting with it directly.

Examples of Recon Data:

- Finding public IP addresses and domains.
- Discovering hidden websites or subdomains.
- Learning about employees via social media (for potential social engineering tests).

STEP 3: SCANNING

Identifying Weaknesses

Scanning involves actively interacting with the target to identify technical details and potential flaws.

- Detecting open network ports.
- Identifying running services and their versions.
- Finding known weak points or misconfigurations in the system.

Goal: To understand exactly how the system works and where it might break.

STEP 4: EXPLOITATION

> Exploitation means actively using a discovered vulnerability to gain unauthorized access to a system.

Important Note:

Ethical hackers execute this phase **only** in fully authorized environments.

STEP 5: REPORTING

The most important deliverable of the assessment.

Explain Findings

Clearly detail what vulnerabilities were found and demonstrate how they could be exploited.

Suggest Solutions

Provide actionable, practical remediation steps to help the organization fix the issues.

Be Professional

A good report must be crystal clear, simple enough for management to understand, and highly professional.

AI IN ETHICAL HACKING

How AI Helps Cybersecurity

Artificial Intelligence is revolutionizing how we defend and test networks.

AI can help with:

- Advanced threat detection.
- Rapid malware analysis.
- Automated, intelligent scanning.
- Detecting anomalous or suspicious activities instantly.

IMPORTANT CYBERSECURITY TOOLS

Nmap Used primarily for network discovery and security auditing (Scanning phase).

Wireshark A powerful network protocol analyzer used to capture and read data packets.

Burp Suite The leading toolkit used for testing the security of web applications.

Metasploit A widely used penetration testing framework that makes hacking simpler.

Kali Linux A specialized operating system packed with hundreds of pre-installed security tools.

CAREERS IN CYBERSECURITY

Possible Career Paths

The field is vast and offers numerous specializations:

- Ethical Hacker / Penetration Tester
- Security Analyst / SOC Analyst
- Digital Forensics Expert
- Security Engineer
- Security Consultant

The Good News

Cybersecurity jobs are growing rapidly worldwide, offering excellent stability and compensation.

QUESTIONS?

Thank you for your attention.